

High Throughput Sequencing Analysis

Spring 2026

Tentative Syllabus!

Subject to change before start of Spring Semester

Evolving syllabus

Subject to some change during the course of the semester. Under certain circumstances, the instructor may have to alter course requirements, assignment deadlines, and grading procedures; and the university may have to alter the semester calendar.

Course Description

In this course, students will analyze eukaryotic High Throughput Sequencing (aka "Next Generation Sequencing") data. Students will carry out a research project in small groups, have weekly lab meetings, and write lab reports.

Note that this course counts as Writing Intensive (department) but does not count toward Experiential Learning. However, if you do well in the course, absolutely feel free to discuss with the instructor potential research project that might count.

Prerequisites

Students taking this section must have permission of instructor.

Administrative details

Texts:	None needed
Meetings:	Lecture MW 11-11:50pm, WAL-4600
	Workshop Th 9:30-11:20am, Zoom

Contact Information

Instructor:	Michael Osier
Office:	08-1338
Instructor Schedule	<u>Online</u>
Contact:	mvoscl@rit.edu

Topics and Readings

Under certain circumstances, the instructor may have to alter course requirements, assignment deadlines, and grading procedures; and the university may have to alter the academic calendar.

NOTA BENE: Required readings are in myCourses under Content.

		Monday	Wednesday	Thursday		Writing due
Week 1	Jan 12	Introduction, Finding data	HTS basics: technologies and analysis, including readings	Unix/Linux command line		None
Week 2	Jan 19	No class	HTS basics	Trapnell paper	Design analysis flowchart	
Week 3	Jan 26	Assembler readings	Start Trapnell data set - quality control	Project picking, Trapnell QC		First and second picks with justification (due week 4 Monday)
Week 4	Feb 2	RNASeq readings	Trapnell data set - QC debrief and alignment			Weekly reports
Week 5	Feb 9	Composite structures readings	Trapnell data set - Alignment debrief and differential expression			
Week 6	Feb 16	Barcoding readings	First stage analyses - Quality control			
Week 7	Feb 23	eDNA and scRNASeq readings				
Week 8	Mar 2	First stage analyses - Quality control				
Week 9	Mar 9	Spring Break				Optional weekly report
Week 10	Mar 16	Second stage analyses - assembly/alignment	Second stage analyses - assembly/alignment			Weekly report; Required peer evaluations
Week 11	Mar 23	Variant callers and missing genes readings				Weekly reports
Week 12	Mar 30	Third stage analyses - differential expression	Third stage analyses - differential expression and visualization			
Week 13	Apr 6	Long read technologies readings	Third stage analyses - differential expression and visualization			Weekly report; Final paper draft 1 - outline
Week 14	Apr 13	Microbiome and epigenetics readings	Third stage analyses - differential expression and visualization			Weekly report; Final paper draft 2 - half of prose/figures
Week 15	Apr 20	Cleanup/bonus analyses	Cleanup/bonus analyses - SNP calling, create transcriptomes, etc.			Weekly report; Final paper draft 3 - nearly done
Week 16	Apr 27	Cleanup/bonus analyses	Exam week activity			Final paper; Final peer evaluations

Grading

10 weekly lab reports	3 pts each x 10 = 30 pts total
3 Trapnell data set Analyses	10 pts each x 3 = 30 pts
Participation in First stage Analysis	30 pts
Participation in Second stage Analysis	30 pts
Participation in Third stage Analysis	30 pts
3 Final Paper Drafts	2 pts each = 6 pts
Final Paper	24 pts
Two peer evaluations	10 pts each = 20 pts total

Total	200 pts
Bonus: bonus analysis for project	20 pts
Total available bonus	20 pts

A	[190-200]
A-	[180-190)
B+	[173.4-180)
B	[166.7-173.4)
B-	[160-166.7)
C+	[153.4-160)
C	[146.7-153.4)
C-	[140-146.7)
D	[120-140)
F	<120

Projects and grading of progress

Each project will use real HTS data. Student form teams and pick one project during Week 3. Students will spend the rest of the semester analyzing this data set.

For each project, there may be trees of primary, secondary, and tertiary analysis steps, which are dependent on each other. For example, any Secondary Analysis will be dependent upon having a Primary Analysis that was completed correctly. Completion of more than three levels of analysis can result in bonus points! Be sure to ask the instructor well in advance if you wish to carry out the additional work to verify that it is appropriate.

Participation for each step of analysis will be graded for elements such as completeness, choice of analysis method, appropriate parameter choice, demonstration of active participation, and regular online communications through the myCourses Discussion groups. Individual participation, including full participation in all class sessions, will also be observed and incorporated.

Readings

Readings are due before the lecture meetings of that week. These readings are intended both to acclimate you to the topic of the week, and serve as discussion for that week. All papers should be available through myCourses, although some are linked out. *In no case should you have to pay for any articles!*

Lab reports

Over the semester, students must submit individual lab reports for 10 out of the 12 available weeks, counting the project choice. Reports must be 3/4 to 2 single spaced pages for undergraduates, or 1 to 2 pages for graduate students, submitted to the correct myCourses dropbox before the Thursday meeting. Reports must include the following elements:

- What was accomplished since the previous report, especially regarding the previous report's future goals. (Not necessary for first report.)
- Methods used, including commands used with parameters
- What problems you encountered
- Goals for the coming week

Final paper

A final paper written by the group, explaining your work, in a typical journal format, is due submitted to the myCourses dropbox before the end of the last day of classes (Monday, April 27th, 11:59pm). No late papers will be accepted.

Part of a draft, written by the group, is due during Weeks 13 through 15 on the following schedule. Drafts must be submitted to the appropriate myCourses DropBox before Thursday workshop. Each draft is worth 1% of your final grade, for a total of 3%. No late drafts will be accepted for a grade. *For drafts, color your individual text in a color unique to you among your group.* You must use MS Word, LibreOffice, or OpenOffice. Word is available from the COS computer labs and the others are free. Be sure to include one comment at the start of the draft with your full name so that your revision color can be identified.

- Week 13 - Outline of the paper.
- Week 14 - Draft of half of the text.
- Week 15 - Draft of the rest of the paper.

Expectations of the paper are as follows.

- 30-50 pages
 - Minimum of 15 primary sources.
 - No more than half figures/tables. The rest must be prose.
 - 8.5" x 11" paper-size
 - double-spaced
 - Citations do *not* count for or against the page requirements.
 - MS Word 2016 or later, LibreOffice, or OpenOffice. Do not use Google Docs which creates formatting issues.
 - 10 or 12 point Calibri or Times New Roman font.
 - Exactly 1" margins.
 - No more than one page of direct quotes in total.
 - Proper spelling, grammar, and style.
 - Standard scientific sections (Introduction, Materials and Methods, Results, Discussion, References).
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Recording policy

Unless written permission is granted by the faculty member, or a specific accommodation has been approved by the Disability Services Office, students are prohibited from recording lectures or presentations.

Plagiarism/cheating policy

For the first offense, anyone caught plagiarizing or otherwise cheating will receive a 0 (zero) on the assignment, and be referred to the Head of the School of Life Sciences. In the event of a second offense, the student will receive an "F" for the course. Duplicate submissions or excessive patchwriting will also receive a grade of 0 (zero) for the assignment. In the case of especially egregious offenses, the instructor reserves the right to assign a grade of "F" for the course, as per RIT policy.

If you have any questions about whether or not something constitutes plagiarism and/or cheating, please ask the instructor in advance.

Artificial Intelligence (AI) may not be used in this course except for the exploration of topics. Among other things, it may not be used for writing or drafting papers/reports. You may use it to get advice on how to do an analysis, for example. If you have any questions about acceptable use of AI in this course, please email the instructor immediately.

Links

Tools used in class

- [FastQC](#)
- [FastX](#)
- [CutAdapt](#)
- [Bowtie 2](#)

- [Bowtie](#)
- [SAMtools](#)
- [Tophat](#)
- [Cufflinks](#)
- [RSEM](#)
- [GFF3 format](#)
- [SOFA GFF3 format](#)

Other links

[Cornell Bowtie/Cufflinks exercise](#)

Contents last updated 12/12/25

- Changes to the syllabus after the start of the course will be noted here.